Modification of Glial Cell Activation through Dendritic Cell Vaccination: Promises for Treatment of Neurodegenerative Diseases.

PMID: 33713321 | Indexed in PubMed

Accumulation of misfolded tau, amyloid β (Aβ), and alpha-synuclein (α-syn) proteins is the fundamental contributor to many neurodegenerative diseases, namely Parkinson's (PD) and AD. Such protein aggregations trigger activation of immune mechanisms in neuronal and glial, mainly M1-type microglia cells, leading to release of pro-inflammatory mediators, and subsequent neuronal dysfunction and apoptosis. Despite the described neurotoxic features for glial cells, recruitment of peripheral leukocytes to the brain and their conversion to neuroprotective M2-type microglia can mitigate neurodegeneration by clearing extracellular protein accumulations or residues. Based on these observations, it was speculated that Dendritic cell (DC)-based vaccination, by making use of DCs as natural adjuvants, could be used for treatment of neurodegenerative disorders. DCs potentiated by disease-specific antigens can also enhance T helper 2 (Th2)-specific immune response and by production of specific antibodies contribute to clearance of intracellular aggregations, as well as enhancing regulatory T cell response. Thus, enhancement of immune response by DC vaccine therapy can potentially augment glial polarization into the neuroprotective phenotype, enhance antibody production, and at the same time balance neuronal cells' repair, renewal, and protection. The characteristic feature of this method of treatment is to maintain the equilibrium in the immune response rather than targeting a single mediator in the disease and their application in other neurodegenerative diseases should be addressed. However, the safety of these methods should be investigated by clinical trials.

Nuclear respiratory factor-1 (NRF1) induction drives mitochondrial biogenesis and attenuates amyloid beta-induced mitochondrial dysfunction and neurotoxicity.

PMID: 39730291 | Indexed in PubMed

Mitochondrial dysfunction is an important driver of neurodegeneration and synaptic abnormalities in Alzheimer's disease (AD). Amyloid beta (Aβ) in mitochondria leads to increased reactive oxygen species (ROS) production, resulting in a vicious cycle of oxidative stress in coordination with a defective electron transport chain (ETC), decreasing ATP production. AD neurons exhibit impaired mitochondrial dynamics, evidenced by fusion and fission imbalances, increased fragmentation, and deficient mitochondrial biogenesis, contributing to fewer mitochondria in brains of AD patients. Nuclear respiratory factor-1 (NRF1) is a regulator of mitochondrial biogenesis through its activation of mitochondrial transcription factor A (TFAM). Our hypothesis posited that NRF1 induction in neuronal cells exposed to amyloid β1-42 (Aβ1-42) would increase de novo mitochondrial synthesis and improve mitochondrial function, restoring neuronal survival. Following NRF1 messenger RNA (mRNA) transfection of Aβ1-42-treated SH-SY5Y cells, a marked increase in mitochondrial mass was observed. Metabolic programming toward enhanced oxidative phosphorylation resulted in increased ATP production. Oxidative stress in the form of mitochondrial ROS accumulation was reduced and mitochondrial membrane potential preserved. Mitochondrial homeostasis was maintained, evidenced by balanced fusion and fission processes. Ultimately, improvement of mitochondrial function was associated with significant decreases in Aβ1-42-induced neuronal death and neurite disruption. Our findings highlight the potential of NRF1 upregulation to counteract Aβ1-42-associated mitochondrial dysfunction and neurodegenerative cell processes, opening avenues for innovative therapeutic approaches aimed at safeguarding mitochondrial health in AD neurons.

BioASQ Synergy: a dialogue between question-answering systems and biomedical experts for promoting COVID-19 research.

PMID: 39180335 | Indexed in PubMed

This article presents the novel BioASQ Synergy research process which aims to facilitate the interaction between biomedical experts and automated question-answering systems. The proposed research allows systems to provide answers to emerging questions, which in turn are assessed by experts. The assessment of the experts is fed back to the systems, together with new questions. With this iteration, we aim to facilitate the incremental understanding of a developing problem and contribute to solution discovery. The results suggest that the proposed approach can assist researchers to navigate available resources. The experts seem to be very satisfied with the quality of the ideal answers provided by the systems, suggesting that such systems are already useful in answering open research questions. BioASQ Synergy aspires to provide a tool that gives the experts easy and personalized access to the latest findings in a fast-growing corpus of material. In this article, we envisioned BioASQ Synergy as a continuous dialogue between experts and systems to issue open questions. We ran an initial proof-of-concept of the approach, in order to evaluate its usefulness, both from the side of the experts, as well as from the side of the participating systems.

Understanding tumour growth variability in breast cancer xenograft models identifies PARP inhibition resistance biomarkers.

PMID: 39558144 | Indexed in PubMed

Understanding the mechanisms of resistance to PARP inhibitors (PARPi) is a clinical priority, especially in breast cancer. We developed a novel mathematical framework accounting for intrinsic resistance to olaparib, identified by fitting the model to tumour growth metrics from breast cancer patient-derived xenograft (PDX) data. Pre-treatment transcriptomic profiles were used with the calculated resistance to identify baseline biomarkers of resistance, including potential combination targets. The model provided both a classification of responses, as well as a continuous description of resistance, allowing for more robust biomarker associations and capturing the observed variability. Thirty-six resistance gene markers were identified, including multiple homologous recombination repair (HRR) pathway genes. High WEE1 expression was also linked to resistance, highlighting an opportunity for combining PARP and WEE1 inhibitors. This framework facilitates a fully automated way of capturing intrinsic resistance, and accounts for the pharmacological response variability captured within PDX studies and hence provides a precision medicine approach.

SEGCOND predicts putative transcriptional condensate-associated genomic regions by integrating multi-omics data.

PMID: 36394233 | Indexed in PubMed

The compartmentalization of biochemical reactions, involved in the activation of gene expression in the eukaryotic nucleus, leads to the formation of membraneless bodies through liquid-liquid phase

separation. These formations, called transcriptional condensates, appear to play important roles in gene regulation as they are assembled through the association of multiple enhancer regions in 3D genomic space. To date, we are still lacking efficient computational methodologies to identify the regions responsible for the formation of such condensates, based on genomic and conformational data. In this work, we present SEGCOND, a computational framework aiming to highlight genomic regions involved in the formation of transcriptional condensates. SEGCOND is flexible in combining multiple genomic datasets related to enhancer activity and chromatin accessibility, to perform a genome segmentation. It then uses this segmentation for the detection of highly transcriptionally active regions of the genome. At a final step, and through the integration of Hi-C data, it identifies regions of putative transcriptional condensates (PTCs) as genomic domains where multiple enhancer elements coalesce in 3D space. SEGCOND identifies a subset of enhancer segments with increased transcriptional activity. PTCs are also found to significantly overlap highly interconnected enhancer elements and super enhancers obtained through two independent approaches. Application of SEGCOND on data from a well-defined system of B-cell to macrophage transdifferentiation leads to the identification of previously unreported genes with a likely role in the process. Source code and details for the implementation of SEGCOND is available at https://github.com/AntonisK95/SEGCOND. Supplementary data are available at Bioinformatics online.

Identification of blood transcriptome modules associated with suicidal ideation in patients with major depressive disorder.

PMID: 39774242 | Indexed in PubMed

The risk of suicide in patients with major depressive disorder (MDD) poses a major concern, with studies suggesting that genetics may be a contributing factor. Although there are many transcriptomic studies on postmortem brain tissue related to suicidal behavior, the blood transcriptional mechanisms of suicidal ideation (SI) remain unknown. This study utilized a weighted gene coexpression network analysis (WGCNA) approach to investigate the associations between gene coexpression modules and SI in individuals with MDD using peripheral blood RNA-seq data from 75 MDD patients with SI (MDD_SI), 82 MDD patients without SI (MDD_nSI), and 149 healthy controls (HC). An ANCOVA was conducted to assess differences in gene coexpression modules among groups, with age and sex included as covariates. The gene ontology (GO) and Kyoto encyclopedia of genes and genomes (KEGG) databases were used to annotate module functions. Results indicated that the magenta module (associated with RNA splicing processes) differentiated MDD_SI from MDD_nSI ($p = 0.021$), while the green module (related to immune and inflammatory responses) distinguished MDD_SI from HC ($p = 0.004$). Additionally, three modules showed differences between MDD_nSI and HC: magenta ($p = 0.009$), brown (related to innate immunity and mitochondrial metabolism; $p = 0.001$), and turquoise (associated with energy metabolism and neurodegeneration; $p = 0.005$). Our findings highlight that gene expression regulation, immune response, and inflammation may be linked to SI in patients with MDD, while pathways associated with innate immunity, energy metabolism, mitochondrial function, and neurodegeneration appear to be more broadly related to MDD.

Genetic correlations between suicide attempts and psychiatric and intermediate phenotypes adjusting for mental disorders.

PMID: 37559484 | Indexed in PubMed

Suicide attempts are a moderately heritable trait, and genetic correlations with psychiatric and related intermediate phenotypes have been reported. However, as several mental disorders as well as major depressive disorder (MDD) are strongly associated with suicide attempts, these genetic correlations could be mediated by psychiatric disorders. Here, we investigated genetic correlations of suicide attempts with psychiatric and related intermediate phenotypes, with and without adjusting for mental disorders. To investigate the genetic correlations, we utilized large-scale genome-wide association study summary statistics for suicide attempts (with and without adjusting for mental disorders), nine psychiatric disorders, and 15 intermediate phenotypes. Without adjusting for mental disorders, suicide attempts had significant positive genetic correlations with risks of attention-deficit/hyperactivity disorder, schizophrenia, bipolar disorder, MDD, anxiety disorders and posttraumatic stress disorder; higher risk tolerance; earlier age at first sexual intercourse, at first birth and at menopause; higher parity; lower childhood IQ, educational attainment and cognitive ability; and lower smoking cessation. After adjusting for mental disorders, suicide attempts had significant positive genetic correlations with the risk of MDD; earlier age at first sexual intercourse, at first birth and at menopause; and lower educational attainment. After adjusting for mental disorders, most of the genetic correlations with psychiatric disorders were decreased, while several genetic correlations with intermediate phenotypes were increased. These findings highlight the importance of considering mental disorders in the analysis of genetic correlations related to suicide attempts and suggest that susceptibility to MDD, reproductive behaviors, and lower educational levels share a genetic basis with suicide attempts after adjusting for mental disorders.

What can psychiatric genetics offer suicidology?

PMID: 11727895 | Indexed in PubMed

There is good evidence from recent studies that depression is familial, and that a substantial proportion of the variation in liability is explained by genes. Suicidal behavior, including completed suicide, also seems to cluster in families. First-degree relatives of individuals who have committed suicide (included dizygotic twins) have more than twice the risk of the general population. For identical co-twins of suicides, the relative risk increases to about 11. Applying a simple structural equation model to the published data suggests a heritability for completed suicide of about 43% (95% confidence intervals 25-60). It is not known at present whether the genes predisposing to suicide are identical with those predisposing to affective disorder, but since only about half of those committing suicide have a diagnosis of depression, it seems probable that the overlap is incomplete. The mode of inheritance of suicidal behavior is almost certain to be complex, involving many genes. There have already been some initial studies of allelic association with polymorphisms in candidate genes such as those involved in serotonergic transmission. Further progress is likely to come from candidate gene and linkage disequilibrium studies that are capable of detecting multiple genes of small effect.

Genetics of suicide: an overview.

PMID: 14965851 | Indexed in PubMed

Risk for suicide may have heritable contributions. Evidence supporting this hypothesis includes strong and consistent findings from more than 20 controlled family studies indicating nearly 5-fold greater relative risk of suicidal acts among relatives of index cases with suicidal behavior compared to relatives of nonsuicidal controls. Relative risk was greater for completed suicide than for attempts. Contributions of genetic instead of environmental factors are indicated by a higher average concordance for suicidal behavior among co-twins of suicidal identical twins compared to fraternal twins or to relatives of other suicidal subjects, in at least seven studies. Three studies indicate significantly greater suicidal risk, particularly for completed suicide, among biological versus adoptive relatives of suicidal or mentally ill persons adopted early in life. Molecular genetics studies have searched inconclusively for associations of suicidal behavior with genes mainly for proteins required for central serotonergic neurotransmission. Complex interactions of environmental with heritable risk and protective factors for suicide and psychiatric illnesses or vulnerability traits are suspected, but specific intervening mechanisms remain elusive. Familial or genetic risks for psychiatric factors strongly associated with suicide, such as major affective illnesses and alcohol abuse, as well as impulsive or aggressive traits, have not consistently been separated from suicidal risk itself.

[The genetics of suicidal behavior].

PMID: 19306130 | Indexed in PubMed

There is consistent evidence suggesting that genetic factors play an important role in predisposition to suicidal behavior. Family, twin, and adoption studies have demonstrated that there is a genetic dimension to suicide. Although there is some overlap between suicide and mood disorders, even among psychiatric groups with the highest risk, some patients never attempt suicide, indicating the importance of a diathesis or genetic link to suicide that is independent of the underlying psychiatric disorder. Over the last 3 decades research has shown that there is a relationship between suicide, aggressiveness, and impulsivity. It is possible that genetic factors may be related to personality traits such as impulsiveness and aggressiveness, which in turn may lead to suicide attempts. An increasing number of molecular genetic studies have been carried out among cases involving suicidal behavior and candidate genes thought to be related to suicide. The most important candidate genes include the serotonin transporters (SERTs), tryptophan hydroxylase (TPH), some serotonin receptors (5HT1A, 5HT1B, and 5HT2A), catechol-O-methyltransferase (COMT), monoamine oxidase-A (MAO-A), and tyrosine hydroxylase (TH). The aim of this review was to assess the genetic dimension of suicide behavior, both current and historical.